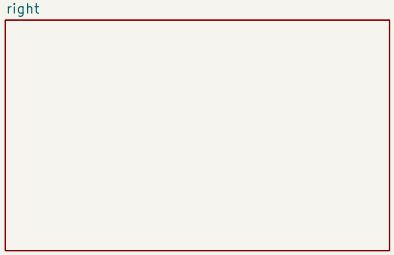
Keygen: one PCB carrying both halves joined by snap-off tabs.
Left half = references 1xx, right half = 2xx. Each half has its own nets (separate GND, VCC).

left



File: left.kicad_sch

right



File: right.kicad_sch

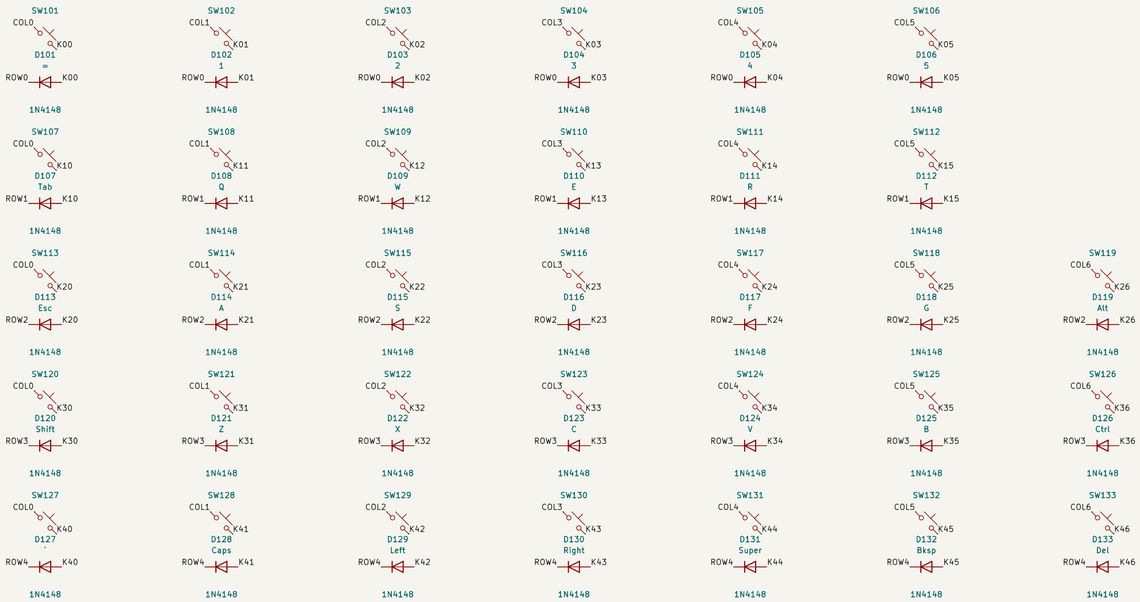
Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen

Sheet: /
File: keygen.kicad_sch

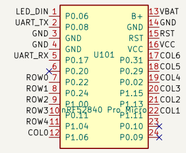
Title: Keygen

Size: A4	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/3

KEY MATRIX (COL2ROW: switch to COLn, diode cathode to ROWn)

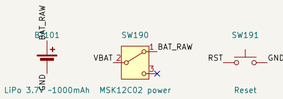


CONTROLLER (mounted on the PCB underside, USB-C to the top edge)



UART: tip = P0.08, ring1 = P0.17 on BOTH halves.
Left firmware: tx P0.08 / rx P0.17.
Right firmware: tx P0.17 / rx P0.08 (crossed in software).
Spare: P0.20, P0.09, P0.10

POWER



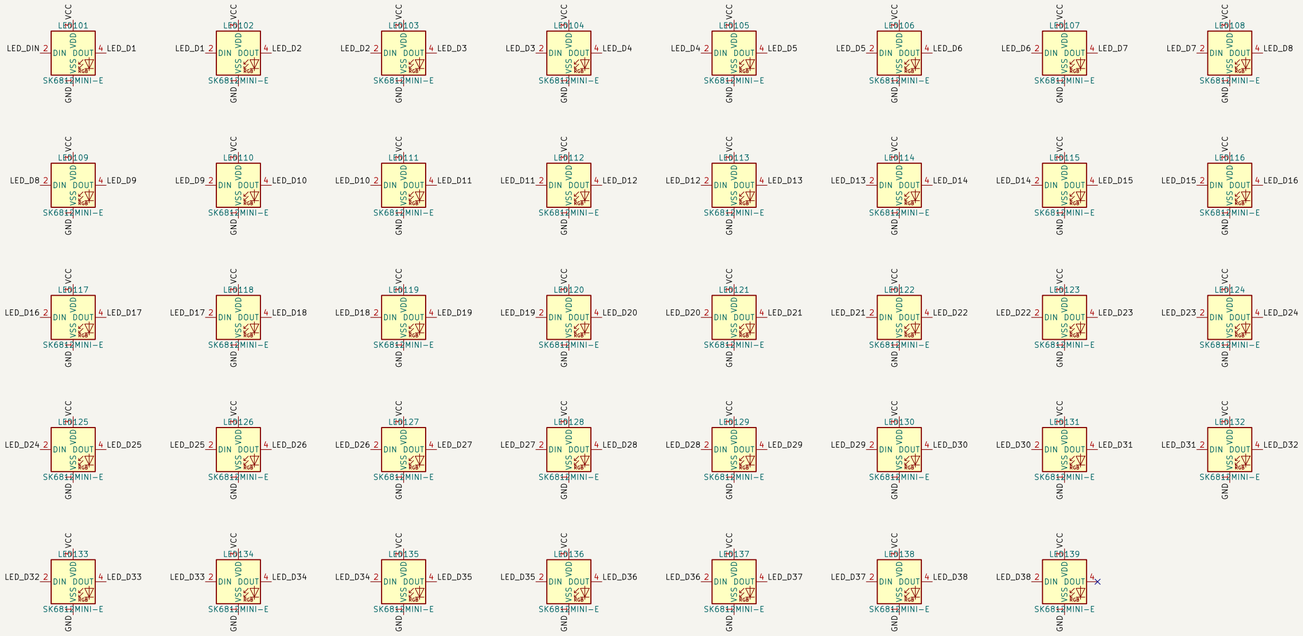


C1/C2 sit next to the first LED of the chain.

TRRS (optional wired split; GND/TX/RX only, never battery power)



PER-KEY + UNDERGLOW LEDs (33 per-key, 6 underglow) — powered from the controller's switched VCC (P0.13); cap brightness in firmware



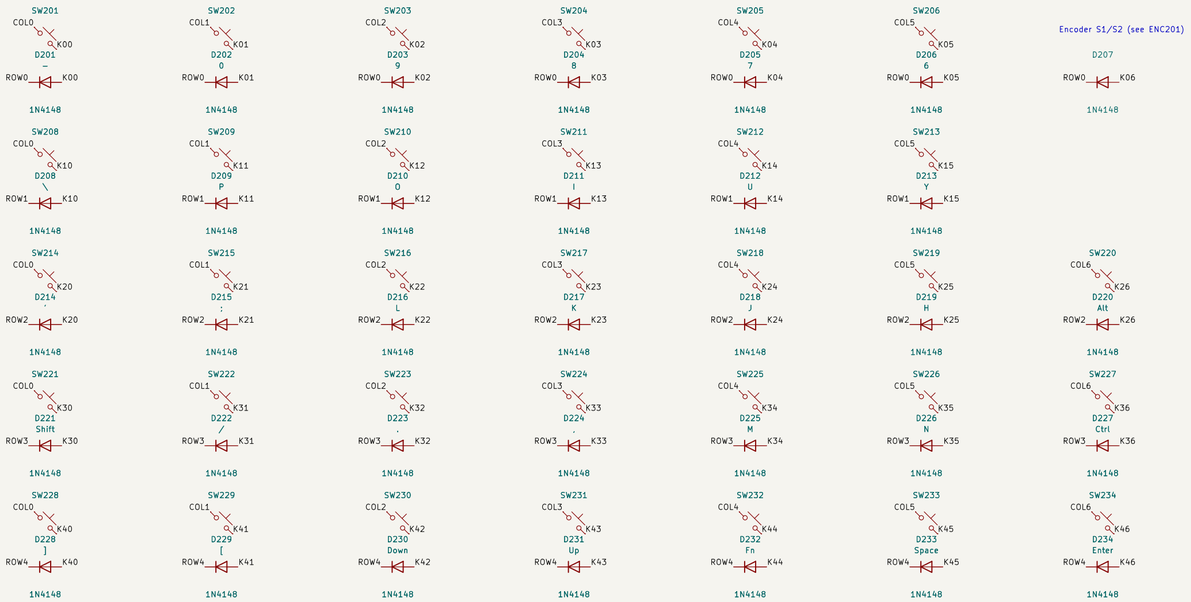
Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen

Sheet: /left/
File: left.kicad_sch

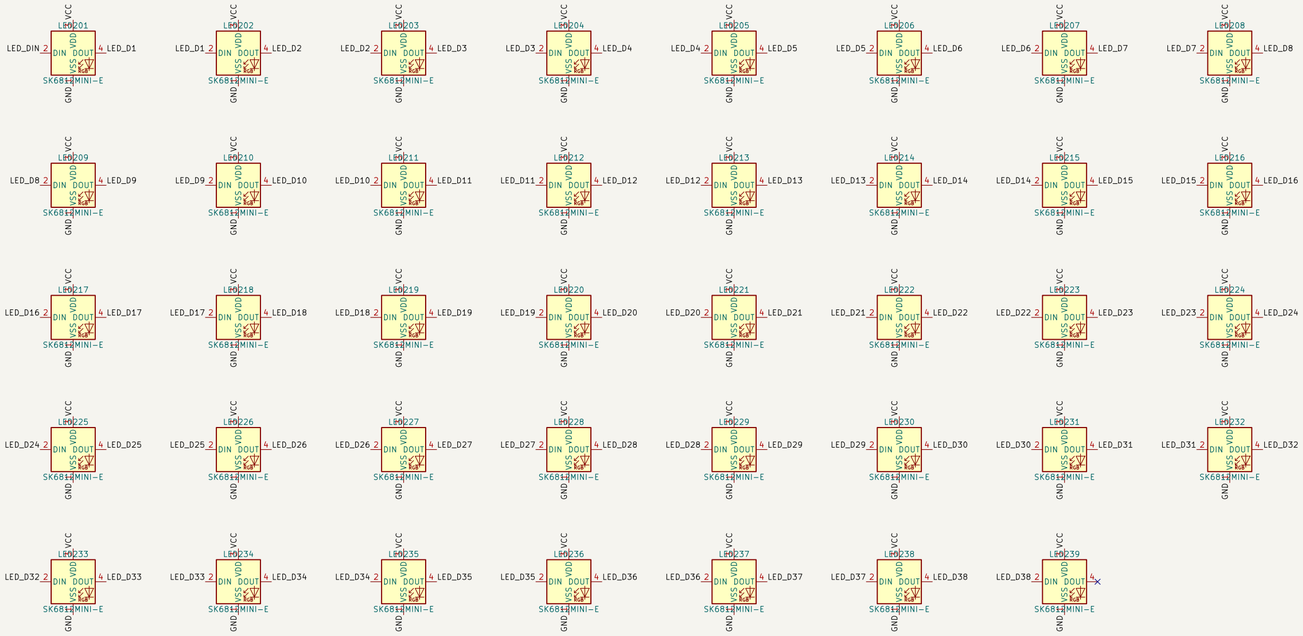
Title: Keygen — left half

Size: A2	Date: 2026-10-19	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 2/3

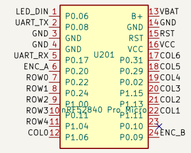
KEY MATRIX (COL2ROW: switch to COLn, diode cathode to ROWn)



PER-KEY + UNDERGLOW LEDs (33 per-key, 6 underglow) — powered from the controller's switched VCC (P0.13); cap brightness in firmware

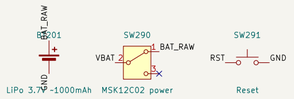


CONTROLLER (mounted on the PCB underside, USB-C to the top edge)



UART: tip = P0.08, ring1 = P0.17 on BOTH halves.
Left firmware: tx P0.08 / rx P0.17.
Right firmware: tx P0.17 / rx P0.08 (crossed in software).
Encoder B on P0.09 (NFC pin): set nfct-pins-as-gpios.
Spare: P0.10

POWER



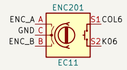


C1/C2 sit next to the first LED of the chain.

TRRS (optional wired split; GND/TX/RX only, never battery power)



ROTARY ENCODER (EC11) — push switch in the matrix at ROW0/COL6



Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen
Sheet: /right/
File: right.kicad_sch

Title: Keygen — right half

Size: A2	Date: 2026-10-19	Rev: 0.1
KiCad E.D.A. 10.0.7		Id: 3/3